Min Number Of Jumps

AlgoExpert · Hard · Dynamic Programming

The problem

- Each value is the max jump length from that index; fewest jumps to reach the end.
- Greedy BFS-by-ranges.

Intuition

- Think in BFS levels: each jump extends the farthest reachable index.

Brute force vs optimal

- DP: for each i, check all earlier j → $O(n^2)$
- Greedy BFS by ranges → O(n)
- Track current range end and farthest reach

Approach

- 1. Track the farthest reach and current jump end
- 2. When we pass the jump end, take another jump
- 3. Count jumps until the end

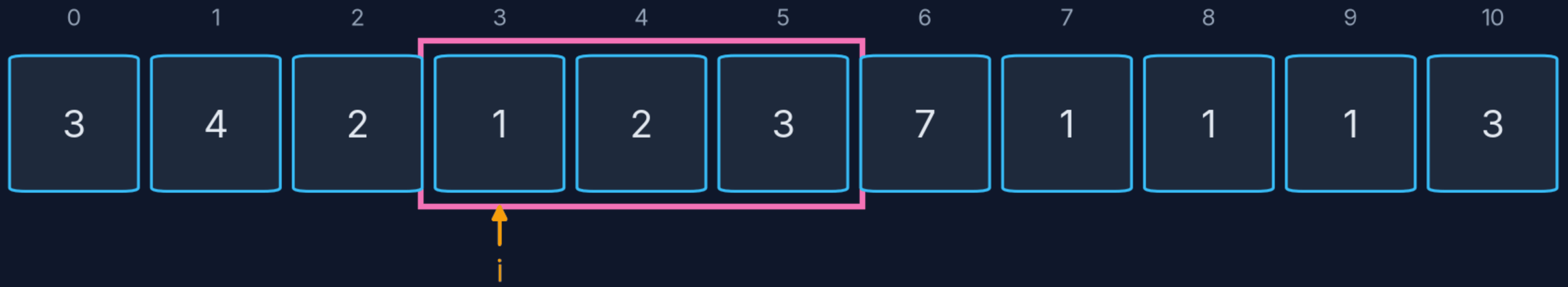
Worked example




```
reached the end of this jump's range → jump #1
new range ends at 3
```

Jump 1: reach up to index 3

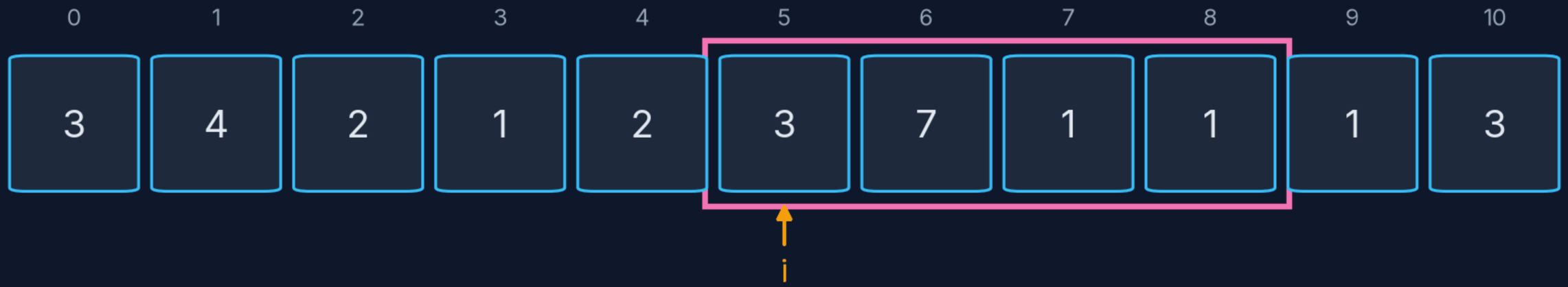
Worked example




```
reached the end of this jump's range → jump #2
new range ends at 5
```

Jump 2: reach up to index 5

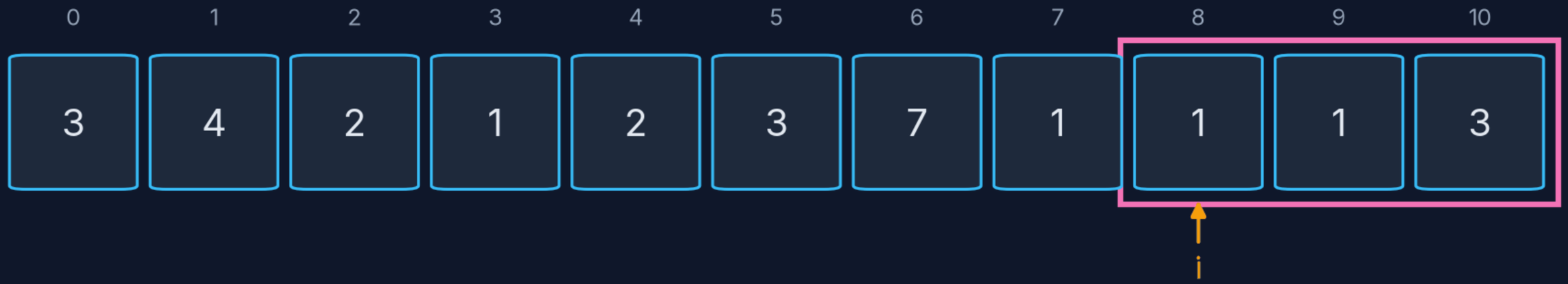
Worked example




```
reached the end of this jump's range → jump #3
new range ends at 8
```

Jump 3: reach up to index 8

Worked example




```
reached the end of this jump's range → jump #4
new range ends at 13
```

Jump 4: reach up to index 13

Java solution

```java
public class MinNumberOfJumps {
    public static int minNumberOfJumps(int[] a) {
        int jumps = 0, curEnd = 0, farthest = 0;
        for (int i = 0; i < a.length - 1; i++) {
            farthest = Math.max(farthest, i + a[i]);
            if (i == curEnd) { jumps++; curEnd = farthest; }
        }
        return jumps;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Don't process the last index
- Single element → 0
- Always reachable (given)

Interview tips & follow-ups

- Explain the BFS-levels view

Recap

- Think in BFS levels: each jump extends the farthest reachable index.
- Complexity: O(n)